Internship Report M2

Exploratory modelling of collective effects in the FCC-ee High-Energy Booster using machine learning

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Abstract

Collective effects in modern particle accelerators generate nonlinear correlations in the six-dimensional beam phase space and require computationally expensive multiparticle simulations. This work develops exploratory surrogate models for the evolution of beam distributions in the High-Energy Booster of the Future Circular Collider (FCC-ee), with the objective of accelerating exploratory studies while retaining physically relevant distributional information.

Two complementary modelling strategies are developed. Reduced longitudinal dynamics are learned directly as mappings between density functions using Fourier Neural Operators. For the six-dimensional problem, beam states are represented by the fifteen pairwise two-dimensional phase-space marginals, supplemented by their centroids and RMS sizes. A conditional variational autoencoder compresses these projected distributions into a probabilistic latent representation, whose evolution through the accelerator lattice is modelled using a causal Transformer, a local-window Transformer, or a Graph Neural Operator.

The comparison of the latent propagators shows that restricting the support of information transfer provides a beneficial inductive bias for the investigated transport problem. Relative to global causal attention, local-window attention reduces distributional reconstruction errors and systematic artefacts. The Graph Neural Operator achieves the lowest relative histogram error, decreasing it from 0.681 for the global Transformer to 0.511, while requiring approximately 2.34 ms per sample under the reported evaluation conditions, roughly one eighth of the 18.74 ms global-Transformer latency. The windowed Transformer and Graph Neural Operator exhibit complementary performance across pointwise and transport-based distributional metrics.

These results demonstrate the potential of structured latent operator models as fast surrogates for simulated beam-distribution evolution. The conclusions remain restricted to the sampled simulation domain and to the fifteen-marginal representation, which does not uniquely determine the complete six-dimensional joint distribution. Nevertheless, the study identifies structured causal support, explicit physical-statistics prediction, and graph-based latent propagation as promising components for future uncertainty-aware accelerator surrogate models.

Keywords

Machine learning, accelerator physics, FCC-ee, collective effects, beam dynamics, neural operators, Fourier Neural Operator, conditional variational autoencoder, Graph Neural Operator, Transformer, surrogate modelling.

1 Introduction

1.1 Motivation

Collective effects are among the principal performance-limiting phenomena in modern circular accelerators. As beam intensity increases, the approximation of independent particles evolving only under externally imposed electromagnetic fields becomes insufficient. The beam interacts with itself and with its environment through wakefields, beam-coupling impedance, space charge, synchrotron radiation, and other collective mechanisms. These interactions may produce emittance growth, coherent tune shifts, bunch lengthening, coupled-bunch motion, transverse mode coupling, and single-bunch instabilities. Wakefield and impedance formalisms therefore provide the standard language for describing beam–environment interactions and evaluating stability margins [1].

These questions are particularly relevant to the Future Circular Collider for electrons and positrons (FCC-ee). Within the proposed injector chain, the High-Energy Booster (HEB) must accumulate and ramp the beam while preserving its phase-space quality. At injection, the interplay between the lattice, vacuum-chamber properties, wakefields, resistive-wall impedance, radiation, and multi-bunch motion motivates computational studies over a broad parameter domain [2]. High-fidelity multiparticle tracking remains indispensable for such studies, but repeated simulations make exhaustive scans and uncertainty studies expensive.

Machine-learning surrogates offer a complementary approach: after training on reference simulations, they can provide inexpensive evaluations for previously unseen configurations inside the sampled domain. The objective is not to replace tracking codes or to claim machine validity beyond the training distribution. It is to determine which representations and architectural biases permit fast approximation of simulated beam-distribution evolution while retaining the observables needed for exploratory analysis. Related machine-learning applications in accelerator physics include diagnostics, correction, optimisation, and simulation surrogates [3, 4].

1.2 Scope and problem statement

In canonical order, the phase-space state is written as

$$\xi_{\text{can}} = (x, p_x, y, p_y, \zeta, \delta) \in \mathbb{R}^6. \tag{1}$$

Let $\rho_s(\xi)$ denote the beam density at longitudinal position, lattice index, or turn s. For fixed machine and collective-effect parameters μ, the target evolution is the generally nonlinear operator

$$\mathcal{G}_{\Delta s,\mu} : \rho_s \longmapsto \rho_{s+\Delta s}. \tag{2}$$

Without collective forces, this operator reduces to the pushforward induced by a single-particle lattice map. With self-consistent forces, the map depends on the distribution itself.

The full six-dimensional joint density is not reconstructed uniquely in this work. Instead, each simulated beam state is represented by its fifteen pairwise two-dimensional marginals, together with six centroids and six RMS sizes. To match the numerical implementation and the plotted pair indices, histogram channels use the storage order

$$\xi_{\text{data}} = (x, y, \zeta, p_x, p_y, \delta), \tag{3}$$

which is a fixed permutation of Eq. (1). This distinction is maintained throughout the report: physical equations use canonical order, whereas pair indices and figure labels use the data order.

The central research question is therefore:

> Can simulated beam evolution be approximated by learned operators acting on reduced longitudinal profiles or on a pairwise-marginal representation, and does restricting the support of latent information transfer improve prediction relative to global causal attention?

1.3 Contributions and evaluation strategy

The principal contributions of the internship are:

1. a simulation and preprocessing workflow that converts particle clouds into reduced longitudinal profiles and fifteen pairwise phase-space marginals;
2. a conditional variational autoencoder (CVAE) that compresses the marginal representation while retaining explicit centroids and RMS sizes;
3. a controlled comparison of global causal attention, finite-window causal attention, and Graph Neural Operator (GNO) propagation in the same latent representation;
4. a kernel-support interpretation connecting masked attention and graph message passing at the level of their non-local interaction layers;
5. a joint evaluation using histogram error, transport-based discrepancies, residual bias, physical statistics, and inference time.

The study is exploratory and simulation-based. Its conclusions are restricted to the sampled parameter domain, the available train–validation–test split, and the pairwise-marginal representation. In particular, improved performance of a restricted-support model is interpreted as evidence about the tested surrogate architectures, not as proof that the underlying collective beam dynamics are strictly local.

2 Beam dynamics

2.1 Single-particle maps and distribution transport

In the absence of collective forces, the phase-space coordinates at the entrance of lattice element j are mapped to its exit by

$$\xi_{j+1} = \Phi_{j,\mu_j}(\xi_j), \tag{4}$$

where μ_j contains the local element parameters. In the linear approximation, $\Phi_{j,\mu_j}(\xi) = M_j(\mu_j)\xi$ and a sequence of N elements gives

$$\xi_N = M_{N-1} \cdots M_1 M_0 \xi_0. \tag{5}$$

At distribution level, the corresponding evolution is the pushforward

$$\rho_{j+1} = (\Phi_{j,\mu_j})_{\#}\rho_j, \tag{6}$$

defined weakly, for every suitable test function φ, by

$$\int \varphi(\xi)\ \mathrm{d}\Big(\Phi_{j,\mu_j}\Big)_{\#}\rho_j(\xi) = \int \varphi\Big(\Phi_{j,\mu_j}(\xi)\Big)\ \mathrm{d}\rho_j(\xi). \tag{7}$$

Equation (7) is the definition of the pushforward; it is not, by itself, a statement of phase-space incompressibility. Volume preservation follows when the microscopic map is symplectic, and therefore satisfies $|\det D\Phi_{j,\mu_j}| = 1$ wherever the Jacobian exists.

2.2 Wakefields and impedance

Intense beams interact with surrounding accelerator structures and generate fields that act back on trailing particles. Under the rigid-beam and impulse approximations, the response of the environment to a point-like source is represented by the longitudinal wake $W_\parallel$ and transverse wake $\boldsymbol{W}_\perp$. The longitudinal line density $\lambda(z)$ is normalised such that

$$\int_{-\infty}^{\infty} \lambda(z)\,\mathrm{d}z = 1. \tag{8}$$

Using the convention that positive separation corresponds to a source particle ahead of a trailing test particle, causality is imposed through $W_\parallel(s) = \boldsymbol{W}_\perp(s) = 0$ for $s < 0$. The bunch wake potentials are then

$$V_\parallel[\lambda](z) = \int_{-\infty}^{\infty} W_\parallel(z - z')\lambda(z')\,\mathrm{d}z', \tag{9}$$

$$\boldsymbol{V}_\perp[\lambda](z) = \int_{-\infty}^{\infty} \boldsymbol{W}_\perp(z - z')\lambda(z')\,\mathrm{d}z'. \tag{10}$$

The vanishing of the wake outside its causal support makes the apparently two-sided integral effectively one-sided. These convolution laws motivate an operator-learning description: the input function λ is mapped to an induced field through a structured integral kernel [1].

The same response is described spectrally by the beam-coupling impedance. With the Fourier convention used in this report,

$$Z_{\parallel}(\omega) = \frac{1}{c}\int_{-\infty}^{\infty} W_{\parallel}(z)e^{i\omega z/c}\,\mathrm{d}z, \tag{11}$$

$$\boldsymbol{Z}_{\perp}(\omega) = \frac{i}{c}\int_{-\infty}^{\infty} \boldsymbol{W}_{\perp}(z)e^{i\omega z/c}\,\mathrm{d}z. \tag{12}$$

Alternative sign conventions are possible and must be used consistently with the longitudinal coordinate. For real wake functions, $Z^*(\omega) = Z(-\omega)$. The Panofsky–Wenzel relation connects longitudinal and transverse wakes,

$$\nabla_{\perp}W_{\parallel}(z, \boldsymbol{r}_{\perp}) = \frac{\partial}{\partial z}\boldsymbol{W}_{\perp}(z, \boldsymbol{r}_{\perp}), \tag{13}$$

and provides a structural consistency check for impedance calculations [1, 5].

2.3 Longitudinal kinetic dynamics

Let z denote the longitudinal displacement relative to the synchronous particle and δ the relative energy deviation. With the prime denoting differentiation with respect to the chosen longitudinal evolution coordinate s, a reduced conservative model is

$$z' = -\eta\delta, \qquad \delta' = K(z), \tag{14}$$

where η is the phase-slip factor and K is the total longitudinal restoring force in the adopted normalisation. The corresponding density $\psi(z, \delta, s)$ satisfies

$$\frac{\partial\psi}{\partial s} - \eta\delta\frac{\partial\psi}{\partial z} + K(z)\frac{\partial\psi}{\partial\delta} = 0, \tag{15}$$

or equivalently $\partial_s\psi + \{\psi, H\} = 0$. When synchrotron-radiation damping and quantum excitation are included, a reduced Vlasov–Fokker–Planck model takes the form

$$\frac{\partial\psi}{\partial s} - \eta\delta\frac{\partial\psi}{\partial z} + K(z)\frac{\partial\psi}{\partial\delta} = \frac{2}{c\tau_z}\frac{\partial}{\partial\delta}\left(\delta\psi + \sigma_\delta^2\frac{\partial\psi}{\partial\delta}\right), \tag{16}$$

where τ_z is the longitudinal damping time and σ_δ the equilibrium energy spread.

Below the microwave-instability threshold, the stationary line density can be expressed through the Haïssinski equation. To avoid using the same symbol for the point-charge wake and the bunch-induced potential, define

$$\mathcal{V}_w[\lambda](z) = \int_{-\infty}^{\infty} W_{\parallel}(z - z')\lambda(z')\,\mathrm{d}z'. \tag{17}$$

The stationary profile satisfies schematically

$$\frac{\mathrm{d}\lambda_h}{\mathrm{d}z} + \left[\frac{z}{\sigma_{z0}^2} - \frac{1}{\eta\sigma_\delta^2}F_h[\lambda_h](z)\right]\lambda_h(z) = 0, \tag{18}$$

with $\int \lambda_h(z)\,\mathrm{d}z = 1$. Equivalently,

$$\lambda_h(z) = A\exp[-V_{\mathrm{eff}}[\lambda_h](z)]\,, \tag{19}$$

where V_{eff} combines the RF potential and the self-induced wake contribution. The dependence of V_{eff} on λ_h makes Eq. (19) a nonlinear fixed-point problem. This is the relation approximated in the reduced longitudinal benchmark; the report does not identify the learned FNO multiplier directly with the physical impedance.

2.4 Distribution-level evolution and modelling scope

For the canonical coordinate vector of Eq. (1), a lattice section with parameters μ defines a microscopic map

$$\Phi_\mu : \mathbb{R}^6 \to \mathbb{R}^6, \qquad \xi_{\text{out}} = \Phi_\mu(\xi_{\text{in}}). \tag{20}$$

When collective effects are included, the distribution-level evolution is generally nonlinear and may be written schematically as

$$\partial_s \rho + \nabla_\xi \cdot (F_{\text{ext}} \rho) + \nabla_\xi \cdot (F_{\text{coll}}[\rho]\rho) = \mathcal{D}[\rho], \tag{21}$$

where F_{ext} is the external lattice force, $F_{\text{coll}}[\rho]$ the density-dependent collective force, and $\mathcal{D}$ represents damping and diffusion.

A formal first-order splitting over a step Δs is

$$\rho_{j+1} \approx e^{\Delta s \mathcal{D}_j} e^{\Delta s \mathcal{C}_j[\rho_j]} (\Phi_{j,\mu_j})_\# \rho_j. \tag{22}$$

This expression is used only to organise the modelling problem into external transport, collective interaction, and dissipative contributions. The present numerical surrogates learn the composite transformations represented in the simulation data; they do not separately identify or enforce the three generators in Eq. (22).

3 Simulation workflow

3.1 Datasets and separation of tasks

Two simulation blocks are used for distinct purposes and are not pooled into a single benchmark. The reduced longitudinal block provides one-dimensional density-to-density examples for the Fourier Neural Operator (FNO). The six-dimensional tracking block provides paired particle clouds for training the CVAE and the latent propagators. Consequently, numerical errors reported for the FNO and for the latent models are task-specific and must not be interpreted as directly comparable scores.

3.2 Six-dimensional Xsuite tracking data

The six-dimensional dataset was generated with an Xsuite-based tracking workflow. For sample s, an initial beam family and a parameter vector $\mu^{(s)}$ are sampled, producing an input cloud

$$Z_{\text{in}}^{(s)} = \left\{ \xi_i^{(s)} \right\}_{i=1}^{N_p}, \tag{23}$$

which is tracked through the selected lattice to obtain

$$Z_{\text{out}}^{(s)} = \Phi_{\mu^{(s)}} \left(Z_{\text{in}}^{(s)} \right). \tag{24}$$

The paired dataset is

$$\mathcal{D}_{\text{track}} = \left\{ \left(Z_{\text{in}}^{(s)}, \mu^{(s)}, Z_{\text{out}}^{(s)} \right) \right\}_{s=1}^{N_s}. \tag{25}$$

The dataset used for the reported six-dimensional figures is

`neural_xsuite_dataset_2026-04-28T09_40_32.npz`.

It contains $N_s = 512$ paired trajectories with $N_p = 4096$ macroparticles per cloud. The beam generator includes Gaussian, correlated, core–halo, mixture, and mismatched families so that the reconstruction task is not restricted to elliptical Gaussian densities. Each state is represented on $N_b = 64$ bins per coordinate.

The split is performed at the paired-trajectory level before initial and final snapshots are concatenated for CVAE pretraining. The resulting subsets contain 410 training pairs, 51 validation pairs, and 51 test pairs. Thus, the two states belonging to one simulated trajectory cannot appear in different subsets. Model selection uses the validation subset; the test subset is reserved for the final comparison.

The parameter vector stored in the dataset is retained as part of every sample. For each component μ_k, the in-distribution domain is defined by the training interval

$$\mathcal{I}_k = \left[\min_{s \in \mathcal{D}_{\text{train}}} \mu_k^{(s)}, \max_{s \in \mathcal{D}_{\text{train}}} \mu_k^{(s)} \right]. \tag{26}$$

No claim of out-of-distribution validity is made outside $\prod_k \mathcal{I}_k$. This definition also prevents information from the validation or test set from enlarging the nominal training domain.

3.3 PyColleff-based longitudinal benchmark

The longitudinal benchmark samples machine and material variables and evaluates a self-consistent response on a uniform grid with $N_\zeta = 512$ points. The conditioning variables include bunch current, chamber radius, and coating thickness. The controlled configuration uses a 3 GeV ring model, a resonator quality factor $Q = 10^4$, three relaxation steps, and under-relaxation coefficient $\alpha = 0.7$. It is used as an operator-learning benchmark and is not presented as a validated end-to-end FCC-ee HEB prediction.

The induced voltage is evaluated spectrally,

$$V_{\text{ind}}(z) = -\mathcal{F}^{-1} \left[Z_\parallel(\omega) \widehat{\lambda}(\omega) \right](z), \tag{27}$$

and combined with the RF voltage. A truncated fixed-point update is

$$\lambda^{(k+1)} = (1-\alpha)\lambda^{(k)} + \alpha \mathcal{T}_\mu \left(\lambda^{(k)} \right), \tag{28}$$

with $\lambda_{\text{out}} = \lambda^{(3)}$. The resulting examples have the form

$$\mathcal{D}_\lambda = \left\{ \left(\lambda_{\text{in}}^{(s)}, \mu^{(s)}, \lambda_{\text{out}}^{(s)} \right) \right\}_{s=1}^{N_\lambda}. \tag{29}$$

3.4 Computational-cost reporting

Reference-simulation wall times were not collected under a single controlled hardware and batching protocol during the exploratory data-generation runs. The empty cost template included in the previous draft has therefore been removed. The report does not claim a measured speed-up relative to Xsuite or PyColleff. The inference times reported in Chapter 5 are used only for comparisons among neural architectures evaluated under the same protocol. A future end-to-end speed-up claim requires measurement of simulation, preprocessing, data-transfer, and inference times on identified hardware.

3.5 Data representation

3.5.1 Empirical measures and reduced profiles

A cloud of N_p macroparticles defines the empirical measure

$$\rho^{N_p} = \frac{1}{N_p} \sum_{i=1}^{N_p} \delta_{\xi_i}. \tag{30}$$

For the longitudinal task, projection onto the fixed grid gives

$$\lambda^h = \Big(\lambda(\zeta_1), \dots, \lambda(\zeta_{N_\zeta})\Big) \in \mathbb{R}^{N_\zeta}. \tag{31}$$

Grid bounds and normalisation constants are estimated from the training data and then frozen for validation and test examples.

3.5.2 Fifteen pairwise phase-space projections

The histogram implementation uses the data order of Eq. (3). Its channel ordering is

$$\begin{aligned}\mathcal{P}_{\text{data}} = \{&(x,y),(x,\zeta),(x,p_x),(x,p_y),(x,\delta),(y,\zeta),(y,p_x),(y,p_y),(y,\delta),\\ &(\zeta,p_x),(\zeta,p_y),(\zeta,\delta),(p_x,p_y),(p_x,\delta),(p_y,\delta)\}.\end{aligned} \tag{32}$$

This definition gives pair indices $p = 0$, 1, 4, and 11 for (x,y), (x,ζ), (x,δ), and (ζ,δ), respectively.

For every pair, fixed bin edges are fitted from the training subset and reused unchanged for validation and test data. This prevents test targets from defining their own coordinate support. The complete representation is

$$\mathcal{H}(\rho) = \{H_p\}_{p=0}^{14} \in \mathbb{R}^{15\times 64\times 64}. \tag{33}$$

Each histogram channel is independently normalised to unit mass. Absolute position and scale are retained through the six centroids and six RMS sizes,

$$m_i = \langle \xi_i \rangle, \qquad \sigma_i = \sqrt{\langle (\xi_i - m_i)^2 \rangle}. \tag{34}$$

Pairwise marginals do not uniquely determine a six-dimensional joint density. Moreover, predicted channels need not be mutually consistent. For coordinate i, let $h_i^{(ij)}$ be the one-dimensional marginal obtained by summing H_{ij} over coordinate j. A diagnostic of shared-marginal consistency is

$$\mathcal{E}_{\text{cons}} = \frac{1}{6} \sum_{i=1}^{6} \frac{1}{|\mathcal{C}_i|} \sum_{(j,k)\in\mathcal{C}_i} \left\| h_i^{(ij)} - h_i^{(ik)} \right\|_1, \tag{35}$$

where $\mathcal{C}_i$ contains pairs of histogram channels sharing coordinate i. This quantity is reported as a diagnostic when available; it is not assumed to vanish by construction.

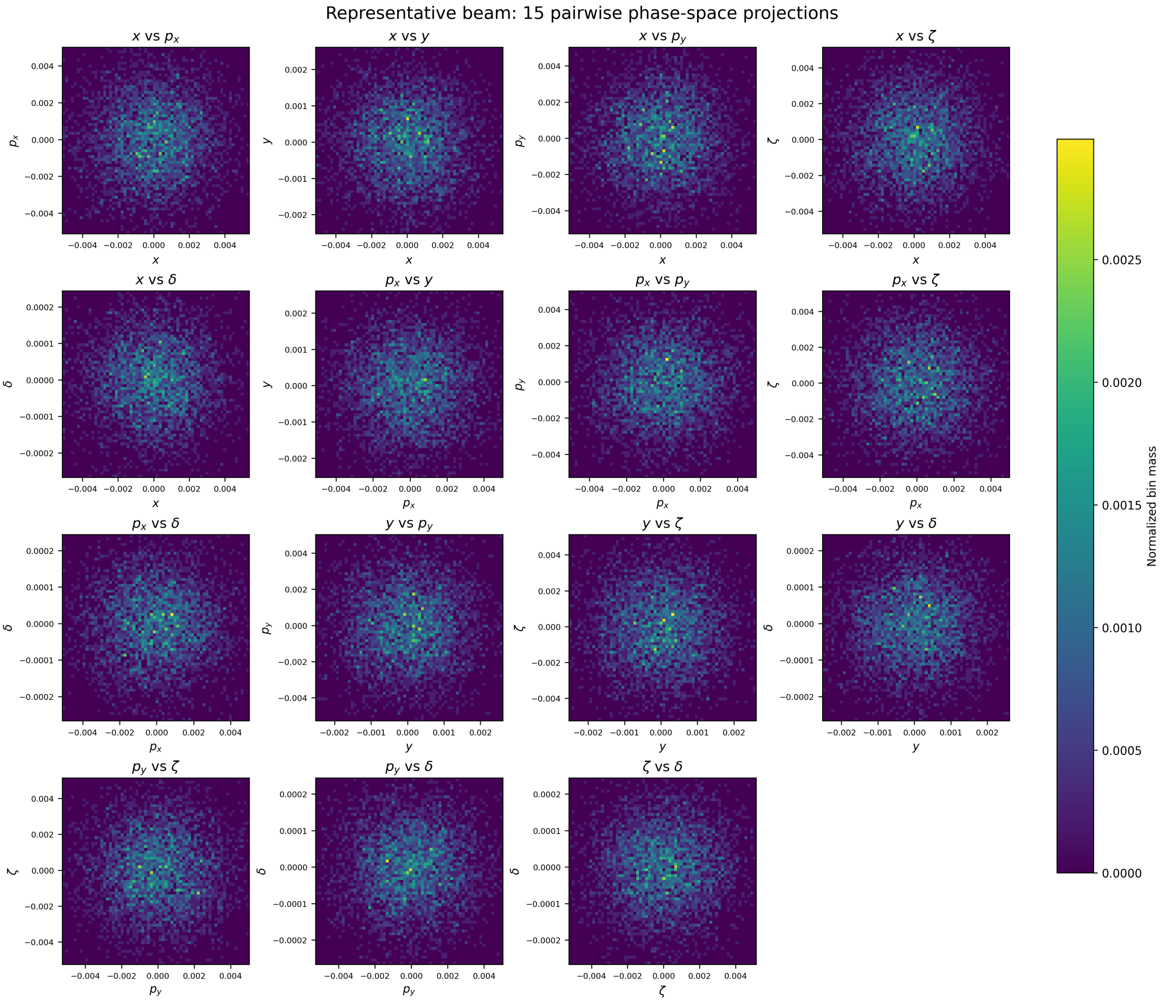

Fig. 1: Representative beam state encoded through its fifteen pairwise two-dimensional phase-space marginals. The associated centroid and RMS vectors retain complementary information about physical position and extension. The figure represents a reduced description of a six-dimensional state, not a unique reconstruction of the complete joint density.

4 Machine-learning methodology

4.1 Overview and comparison principle

The models in this work approximate two related but distinct operators. The longitudinal branch acts directly on a one-dimensional density, whereas the six-dimensional branch acts on the reduced representation defined in Eq. (33). For a dataset

$$\mathcal{D} = \left\{ \left(a^{(s)}, \mu^{(s)}, u^{(s)} \right) \right\}_{s=1}^{N}, \tag{36}$$

the general training objective is

$$\theta^\star = \arg\min_\theta \frac{1}{N}\sum_{s=1}^{N} \ell\Big(\mathcal{G}_\theta(a^{(s)}, \mu^{(s)}), u^{(s)}\Big). \tag{37}$$

The choice of representation and interaction support, rather than generic neural-network depth alone, distinguishes the models compared below.

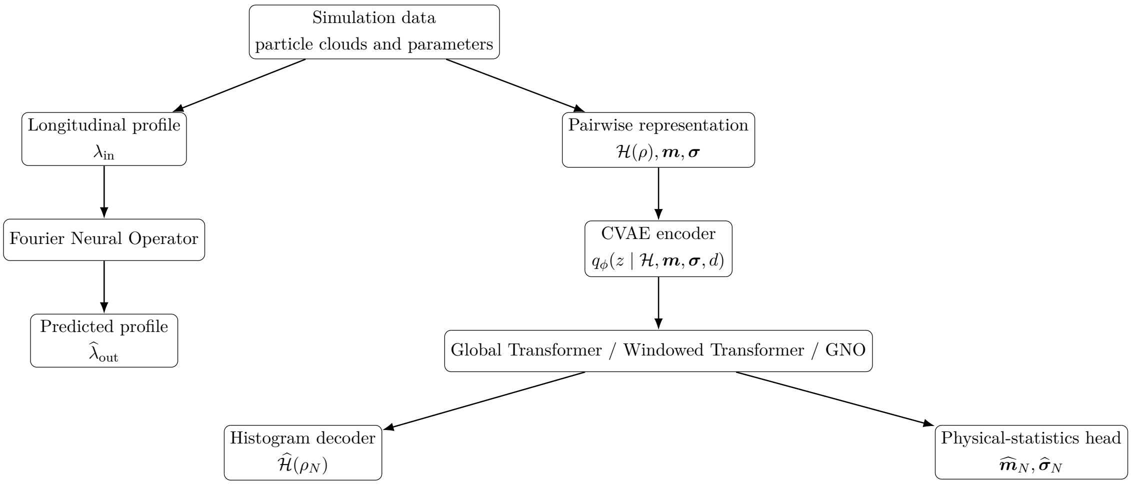

Fig. 2: Implemented modelling hierarchy. The FNO acts on reduced longitudinal profiles. The CVAE compresses the fifteen pairwise marginals; three alternative propagators then evolve the same latent representation. A decoder reconstructs projected density shape, while a separate head predicts six centroids and six RMS sizes.

The comparison between latent propagators uses the same trajectory split, histogram preprocessing, coordinate order, CVAE representation, and output targets. This design reduces, but does not eliminate, confounding from optimisation and random initialisation. The reported comparison corresponds to one selected checkpoint per configuration; repeated-seed uncertainty is therefore identified as a limitation rather than inferred from test-sample variation.

4.2 Fourier Neural Operator

The longitudinal problem is a map between functions sampled on the same uniform grid,

$$\widehat{\mathcal{T}}_\theta : (\lambda_{\text{in}}, \mu) \longmapsto \widehat{\lambda}_{\text{out}}. \tag{38}$$

For a hidden field $v_\ell(z)$, one FNO layer is

$$v_{\ell+1}(z) = \sigma\Big[W_\ell v_\ell(z) + \mathcal{F}^{-1}\left(R_{\theta,\ell}(k)\mathcal{F}[v_\ell](k)\right)(z) + b_\ell(z)\Big], \tag{39}$$

where only a finite set of Fourier modes is retained. The spectral parameterisation is compatible with the convolutional structure of wakefields,

$$V_{\text{ind}}(z) = -\mathcal{F}^{-1}\left[Z_\parallel(\omega)\widehat{\lambda}(\omega)\right](z), \tag{40}$$

but the learned multiplier $R_{\theta,\ell}$ is not interpreted as the physical impedance. It approximates the complete transformation represented in the training data, including RF confinement, relaxation, parameter dependence, and discretisation effects. The study does not experimentally test cross-resolution transfer; the FNO is evaluated on the grid used for training.

4.3 Conditional variational representation

4.3.1 Inputs and conditioning

The CVAE input is the complete channel tensor

$$X = \mathcal{H}(\rho) \in \mathbb{R}^{15\times 64\times 64}. \tag{41}$$

The explicit physical vector contains six coordinate centroids and six RMS sizes in the data order of Eq. (3),

$$a = [\boldsymbol{\sigma}, \boldsymbol{m}] \in \mathbb{R}^{12}. \tag{42}$$

A two-component domain indicator distinguishes input and output snapshots,

$$d_{\text{in}} = (1, 0), \qquad d_{\text{out}} = (0, 1). \tag{43}$$

The domain indicator contains no target value; it only identifies which of the two snapshot domains is being reconstructed. Splitting by trajectory before concatenation prevents it from coupling train and test states.

4.3.2 Encoder, posterior, and decoder

Four residual convolutional blocks map

$$15\times 64\times 64 \longrightarrow 32\times 32\times 32 \longrightarrow 64\times 16\times 16 \longrightarrow 128\times 8\times 8 \longrightarrow 256\times 4\times 4. \tag{44}$$

After flattening, the 4096 convolutional features are concatenated with a and d and projected to $h \in \mathbb{R}^{1024}$. Two heads parameterise a diagonal Gaussian posterior,

$$q_\phi(z \mid X, a, d) = \mathcal{N}\Big(\boldsymbol{\mu}_z, \operatorname{diag}(\boldsymbol{\sigma}_z^2)\Big), \qquad z \in \mathbb{R}^{512}, \tag{45}$$

with

$$z = \boldsymbol{\mu}_z + \boldsymbol{\sigma}_z \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{46}$$

The decoder receives (z, d), but not a. It reconstructs normalised pairwise morphology through four residual upsampling blocks and a final 1×1 convolution. Therefore, the report does not claim that physical scale is supplied directly to the decoder. Instead, scale and position are supervised explicitly through the physical-statistics branch and may also be represented implicitly in z because the encoder used a. This asymmetric conditioning is important when interpreting the latent-correlation diagnostic.

For channel p, non-negativity and unit mass are imposed by

$$\widetilde{H}_p = \operatorname{ReLU}(G_p), \qquad \widehat{H}_p(u, v) = \frac{\widetilde{H}_p(u, v)}{\sum_{u',v'} \widetilde{H}_p(u', v') + \varepsilon}. \tag{47}$$

If the denominator is numerically negligible, a uniform channel is used to avoid division by zero. Unit-mass normalisation enforces mass conservation for each projected probability map, but it does not enforce consistency among channels sharing one coordinate.

4.3.3 Snapshot-pretraining objective

Initial and final snapshots from training trajectories are concatenated only after the split. The implemented loss is

$$\mathcal{L}_{\mathrm{CVAE}} = \mathcal{L}_{\mathrm{rec}}^{\mathrm{in}} + \mathcal{L}_{\mathrm{rec}}^{\mathrm{out}} + \beta \mathcal{L}_{\mathrm{KL}} + \gamma \mathcal{L}_{\sigma} + \delta \mathcal{L}_{m}, \tag{48}$$

with $\beta = 10^{-2}$ and $\gamma = \delta = 10^{-3}$. The KL term is

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{2}\mathbb{E}\left[\sum_{j=1}^{512}\left(\mu_{z,j}^2 + \sigma_{z,j}^2 - 1 - \log \sigma_{z,j}^2\right)\right]. \tag{49}$$

RMS sizes are supervised in logarithmic space to preserve positivity and balance scales,

$$\mathcal{L}_{\sigma} = \mathrm{MSE}\left(\log \widehat{\boldsymbol{\sigma}}, \log \boldsymbol{\sigma}\right), \tag{50}$$

while $\mathcal{L}_m = \mathrm{MSE}(\widehat{\boldsymbol{m}}, \boldsymbol{m})$. Reported KL values are given both as totals and per latent coordinate because the total grows with d_z; only the weighted contribution $\beta\mathcal{L}_{\mathrm{KL}}$ enters the optimisation objective.

4.4 Tokenised latent dynamics

The pretrained encoder maps the initial state to z_0. A propagator then predicts a residual latent evolution conditioned on the ordered lattice tokens,

$$z_{i+1} = z_i + \Delta z_i, \qquad \Delta z_i = f_\theta(z_i, h_i, \mu), \tag{51}$$

where h_i includes element type, length, continuous position, local element parameters, and global conditioning variables. Continuous position is expanded using Fourier features so that the model can represent several longitudinal scales.

4.4.1 Global causal Transformer

For token embeddings $H \in \mathbb{R}^{N \times d}$, one attention head is

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V. \tag{52}$$

The global causal mask permits keys $j \leq i$ and excludes future positions. It therefore preserves directionality but allows direct interaction with the complete upstream sequence [6].

4.4.2 Finite-window causal Transformer

The symbol W denotes the number of accessible tokens, including the current token. The mask is

$$M_{ij}^{(W)} = \begin{cases} 0, & \max(1, i - W + 1) \le j \le i, \\ -\infty, & \text{otherwise.} \end{cases} \tag{53}$$

Thus, $W = 1$ is self-only attention and $W = 4$ gives the four-token band illustrated in Fig. 3. The hidden dimension, heads, feed-forward width, and output projection remain unchanged; the window modifies graph connectivity rather than merely reducing network size. After L layers, information can propagate indirectly over at most approximately $1 + L(W - 1)$ upstream positions in the absence of a global bypass.

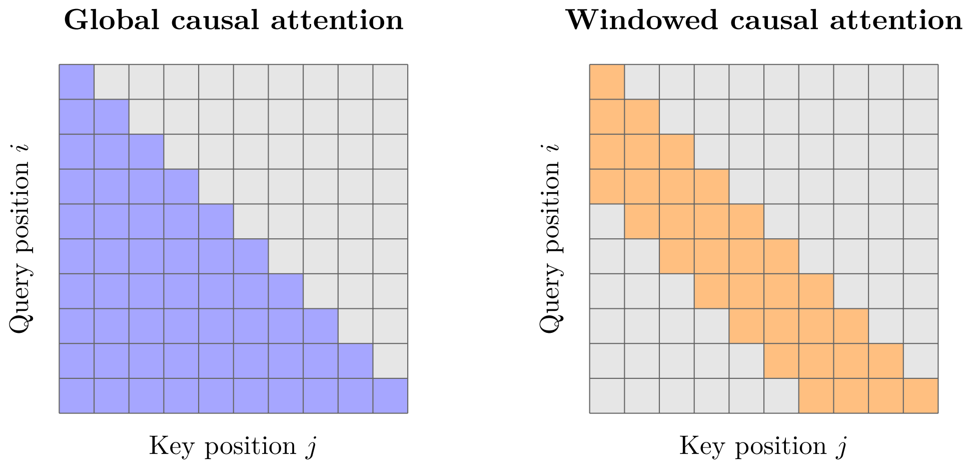

Fig. 3: Support of global causal attention and a four-token causal window. Grey entries are masked. The current token is included in W.

4.4.3 Graph Neural Operator

The lattice is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Nodes correspond to lattice elements or observation positions and incoming edges define the admissible upstream neighbourhood $\mathcal{N}_G(i)$. A latent GNO layer is

$$v_i^{(\ell+1)} = \sigma\left[W_\ell v_i^{(\ell)} + \sum_{j \in \mathcal{N}_G(i)} \omega_{ij} K_{\theta,\ell}\left(h_i, h_j, v_i^{(\ell)}, v_j^{(\ell)}, \mu\right) v_j^{(\ell)}\right]. \tag{54}$$

The implemented kernel receives relative position and element features; four graph-operator layers are used in the reported GNO. The kernel is an effective interaction in latent space and is not identified with a physical wake, transfer matrix, or Green function.

At the non-local interaction level, both windowed attention and the GNO have the discrete form

$$(\mathcal{K}_\Theta v)_i = \sum_{j=1}^{N} A_{ij} K_\Theta(i, j, v_i, v_j, \mu) v_j. \tag{55}$$

For attention, A_{ij} is the causal band and K_Θ is softmax-normalised query–key compatibility. For the GNO, A_{ij} is the directed graph adjacency and K_Θ is an explicitly parameterised graph kernel. This is a computational correspondence, not a proof that the physical dynamics have compact support.

4.5 Physical-statistics head

The histogram decoder predicts normalised distributional shape. A separate head predicts the absolute six-dimensional centroids and RMS sizes,

$$(z_N, \boldsymbol{m}_0, \log \boldsymbol{\sigma}_0, \mu) \longmapsto (\widehat{\boldsymbol{m}}_N, \log \widehat{\boldsymbol{\sigma}}_N). \tag{56}$$

The output has twelve components, not fifteen pair-specific statistics. Pair-specific axes are reconstructed from these six shared coordinate descriptors, which avoids predicting contradictory copies of the same coordinate scale. The head is evaluated independently from the histogram decoder because good normalised shape does not guarantee correct physical position or extension.

4.6 Losses and validation metrics

For a test set of N_{test} paired trajectories and fifteen channels, the mean relative histogram error is

$$\epsilon_{\text{rel}} = \frac{1}{15N_{\text{test}}} \sum_{s=1}^{N_{\text{test}}} \sum_{p=0}^{14} \frac{\left\| \widehat{H}_p^{(s)} - H_p^{(s)} \right\|_2}{\left\| H_p^{(s)} \right\|_2 + \varepsilon}. \tag{57}$$

Residual bias is computed on the same normalised maps,

$$B_H = \left\| \frac{1}{15N_{\text{test}}} \sum_{s,p} \left(\widehat{H}_p^{(s)} - H_p^{(s)} \right) \right\|_2. \tag{58}$$

The Wasserstein-1 distance W_1 and sliced-Wasserstein distance SW are reported separately; they are not interchangeable. Physical-statistics errors are

$$\text{MAE}_m = \frac{1}{6N_{\text{test}}} \sum_{s,d} \left| \widehat{m}_d^{(s)} - m_d^{(s)} \right|, \tag{59}$$

$$\text{MAE}_\sigma = \frac{1}{6N_{\text{test}}} \sum_{s,d} \left| \widehat{\sigma}_d^{(s)} - \sigma_d^{(s)} \right|. \tag{60}$$

For each coordinate, the coefficient of determination is

$$R_d^2 = 1 - \frac{\sum_s (\widehat{q}_d^{(s)} - q_d^{(s)})^2}{\sum_s (q_d^{(s)} - \overline{q}_d)^2}, \qquad q \in \{m, \sigma\}. \tag{61}$$

Because most histogram bins are close to zero, R^2 is not used as the primary density-map metric; it is reserved for the physical-statistics head. Shared-marginal consistency is evaluated with Eq. (35) when the required channel reductions are available.

Tab. 1: Definitions and scope of the principal validation metrics.

Metric	Symbol	Scope
Relative histogram error	ϵ_{rel}	Mean over test samples and 15 channels
Wasserstein-1	W_1	Physical displacement on a projected plane
Sliced Wasserstein	SW	Transport discrepancy from sampled projections
Residual bias norm	B_H	Norm of the mean signed map residual
Centroid/RMS MAE	$\text{MAE}_m, \text{MAE}_\sigma$	Mean over six coordinates
Coordinate score	R_d^2	Physical-statistics head only
Marginal consistency	$\mathcal{E}_{\text{cons}}$	Agreement among channels sharing a coordinate
Inference time	t_{inf}	Neural-model comparison on common hardware

Inference timing excludes simulation and data generation. A controlled simulator speed-up is therefore not reported. The absence of repeated training seeds, parameter-count matching, and a fixed-β_{KL} window ablation is retained explicitly when interpreting the results.

5 Results

5.1 Quantitative overview and scope

Table 2 places the reported experiments at the beginning of the chapter. The FNO, standalone CVAE, and latent propagators solve different tasks, so their errors must not be ranked across blocks. Within the latent-propagator block, all models use the same pairwise representation and held-out test trajectories. A dash denotes a quantity that was not retained under a sufficiently controlled protocol; it is not a value of zero.

Tab. 2: Overview of the models evaluated in this work. The standalone CVAE is a snapshot-reconstruction model. The Transformer and GNO rows compare propagation on the common test set. FNO metrics were not exported under the final common protocol and are therefore treated qualitatively.

Task	Model	ϵ_{rel}	SW	Moment error	B_H	t_{inf} [ms]
Longitudinal	FNO	–	–	–	–	–
Snapshot	CVAE	0.367	0.0096	0.264	–	–
Propagation	Transformer, global	0.681	0.087	9.838	17.337	18.74
Propagation	Transformer, $W = 1$	0.650	0.026	63.047	52.389	14.35
Propagation	Transformer, $W = 8$	0.534	**0.014**	56.169	50.414	12.01
Propagation	CVAE–GNO	**0.511**	0.022	**0.547**	**0.376**	**2.338**

The table exhibits a Pareto trade-off rather than one model winning every metric. The $W = 8$ Transformer has the lowest sliced-Wasserstein discrepancy, whereas the GNO has the lowest relative histogram error, moment error, residual-bias norm, and inference time. Inference timing compares neural models only; Chapter 3 explains why no controlled simulator speed-up is claimed. The table reports selected checkpoints and does not include variation over repeated training seeds.

5.2 Longitudinal operator learning with FNO

The preliminary longitudinal block evaluates the reduced map

$$\widehat{\mathcal{T}}_\theta : (\lambda_{\text{in}}, \mu) \longmapsto \widehat{\lambda}_{\text{out}}. \tag{62}$$

The task provides a controlled demonstration of density-to-density operator learning before the higher-dimensional latent study.

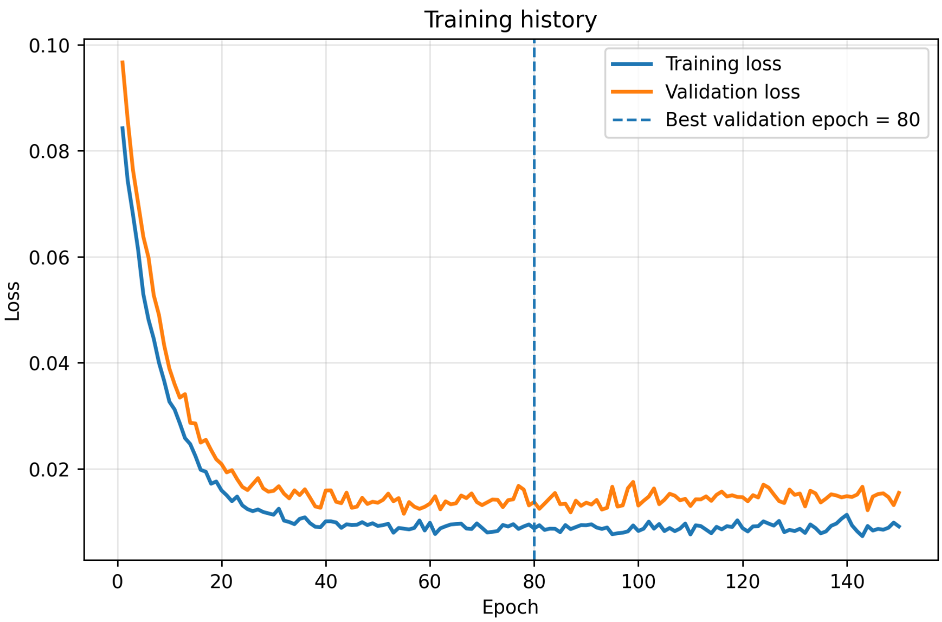

Fig. 4: Training and validation histories for the one-dimensional FNO. The curves document optimisation convergence but do not replace held-out physical metrics.

For the stationary benchmark, the target profile is related to the nonlinear Haïssinski fixed point. Figure 5 illustrates one reconstruction and its signed residual.

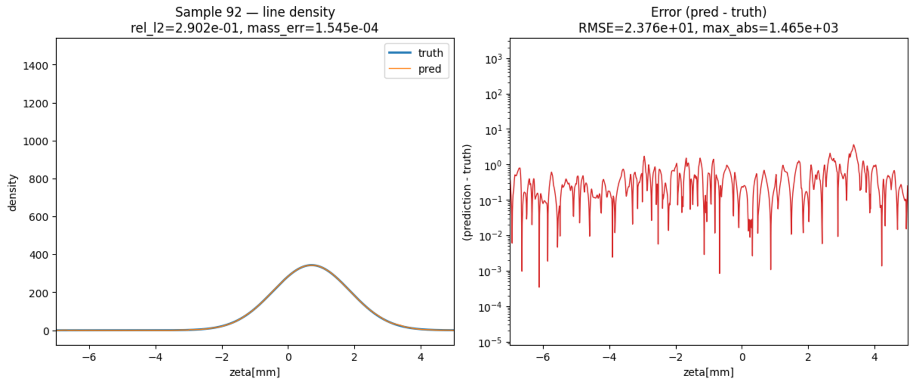

Fig. 5: Illustrative longitudinal reconstruction. The FNO prediction $\widehat{\lambda}_{\text{out}}$ is compared with the reference λ_{out} and with the signed residual $\widehat{\lambda}_{\text{out}} - \lambda_{\text{out}}$.

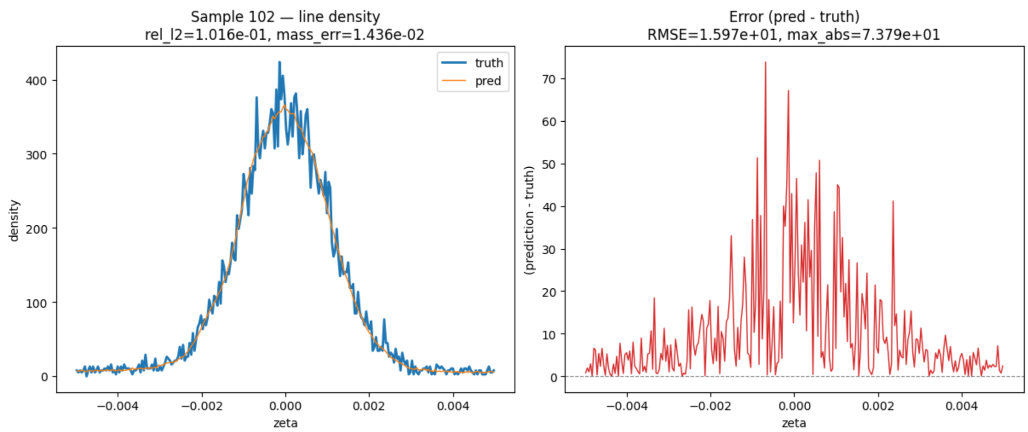

Fig. 6: Second illustrative FNO prediction. A perturbation was applied in the plotting experiment; because no multi-level, repeated-noise study was retained, this figure is not interpreted as evidence of robustness.

No final table containing test-set R^2, centroid error, bunch-length error, Wasserstein distance, and inference time was exported for this preliminary block. The report therefore limits the FNO conclusion to feasibility and does not quantitatively rank it against the six-dimensional models. A robust longitudinal claim requires regeneration of these metrics using the final held-out protocol.

5.3 CVAE snapshot reconstruction

The next block evaluates compression and reconstruction of the fifteen pairwise marginals. Figure 7 shows reference, reconstruction, and residual structure for selected test states. The examples are illustrative; aggregate conclusions are based on Table 3, not on visual selection.

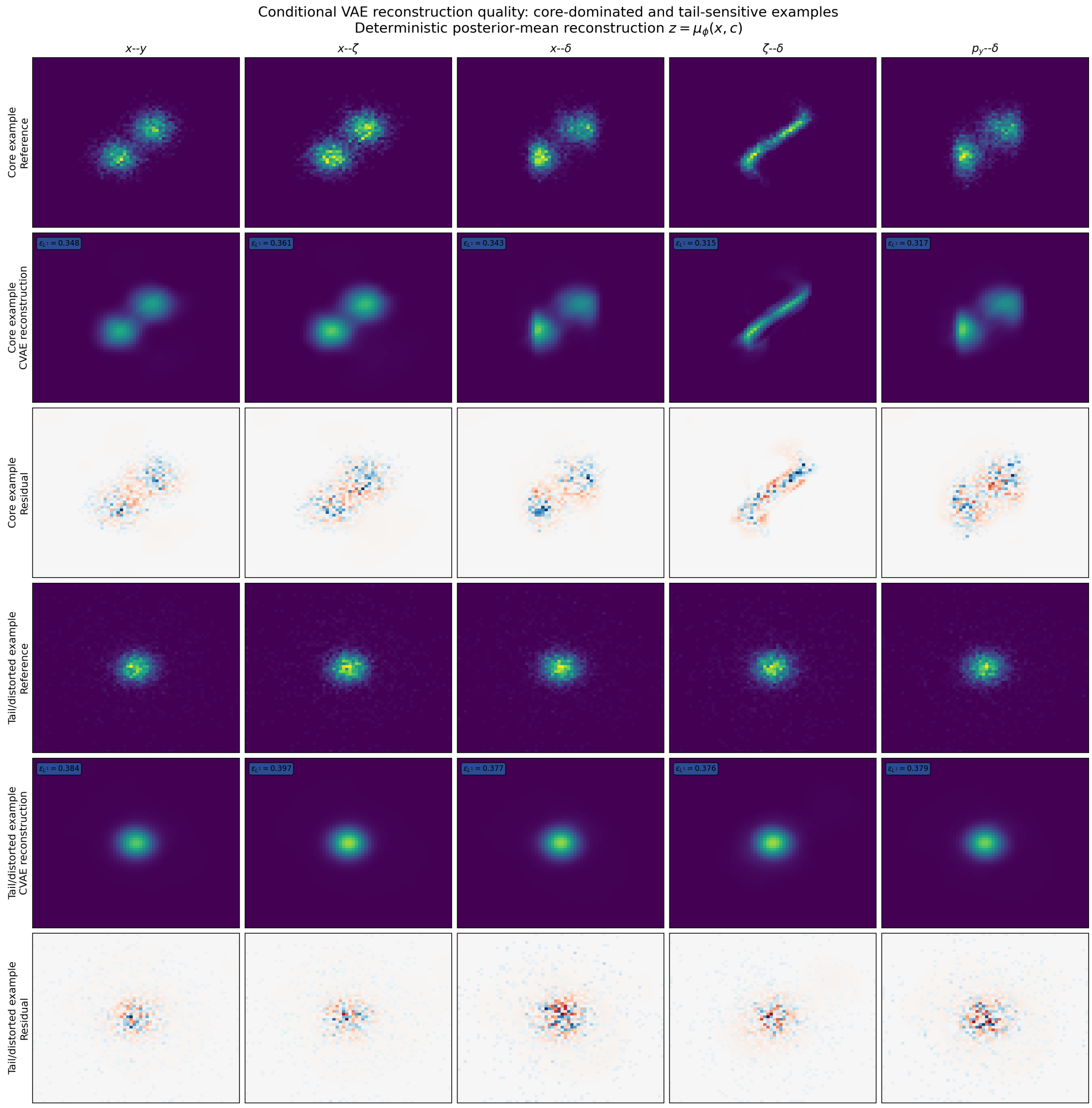

Fig. 7: CVAE reconstruction quality for selected beam states. High-density cores are generally reconstructed more faithfully than low-density tails and distorted structures. Because the complete fifteen-plane panel is visually dense, individual planes should be inspected together with the aggregate test metrics.

Tab. 3: CVAE reconstruction metrics. The KL total is the sum over 512 latent coordinates; the per-coordinate and weighted test values are included to make its scale interpretable.

Metric	Train	Validation	Test
Relative histogram error	0.362	0.357	0.367
Moment error	0.178	0.243	0.264
KL total	929.275	921.513	920.925
KL per latent coordinate	1.815	1.800	1.799
Weighted KL, $\beta\mathcal{L}_{\mathrm{KL}}$	9.293	9.215	9.209
Sliced-Wasserstein distance	0.0092	0.0089	0.0096

The similarity of train, validation, and test histogram errors provides no evidence of a large generalisation gap for this split. However, a relative histogram error of 0.367 is not negligible. The CVAE should therefore be viewed as a lossy representation whose downstream error contains both compression and propagation contributions.

5.3.1 Latent–physical correlation diagnostic

Pearson coefficients were computed between 512 latent coordinates and the twelve supplied physical observables over $N_{\mathrm{val}} = 51$ validation samples,

$$r_{q,z_k} = \frac{\mathrm{Cov}(q, z_k)}{\sigma_q \sigma_{z_k}}. \tag{63}$$

This produces 6144 simultaneous coefficients.

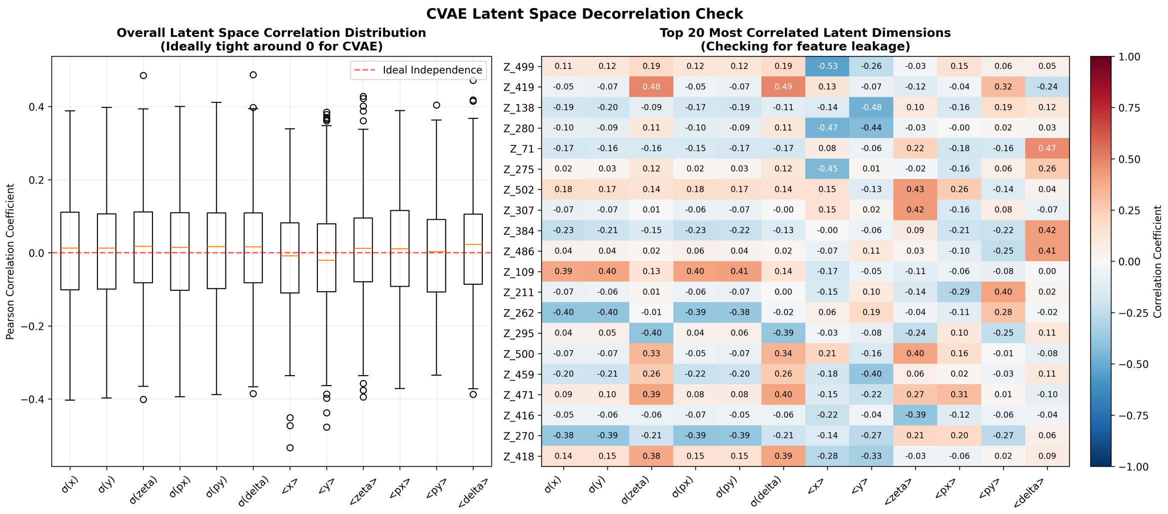

Fig. 8: Pearson correlations between the 512 latent coordinates and six RMS sizes plus six centroids. The figure is a descriptive linear-dependence diagnostic; it is not a test of latent independence or identifiability.

Tab. 4: Global latent–physical Pearson statistics on the validation subset. Threshold counts are descriptive and are not corrected significance tests.

Diagnostic	Value	Fraction of pairs
Mean $\|r\|$	0.1161	–
Median $\|r\|$	0.1006	–
Standard deviation of $\|r\|$	0.0846	–
Maximum $\|r\|$	0.5333	–
$\|r\| \geq 0.30$	198/6144	3.22%
$\|r\| \geq 0.40$	18/6144	0.29%
$\|r\| \geq 0.50$	1/6144	0.016%

The observed coefficients are concentrated near zero, but the combination of $N_{\mathrm{val}} = 51$ and 6144 simultaneous comparisons prevents a strong disentanglement claim. No permutation baseline, multiple-testing correction, out-of-sample linear probe, mutual-information estimate, or HSIC test was retained. The appropriate conclusion is therefore limited: the diagnostic does not reveal widespread large pairwise linear correlations, but it cannot establish statistical independence or exclude nonlinear encoding of scale and position.

5.4 Latent propagation and conditional structure

5.4.1 Global causal attention

The global Transformer permits every token to attend to its complete upstream history. Figures 9 and 10 show illustrative density and conditional-mean diagnostics.

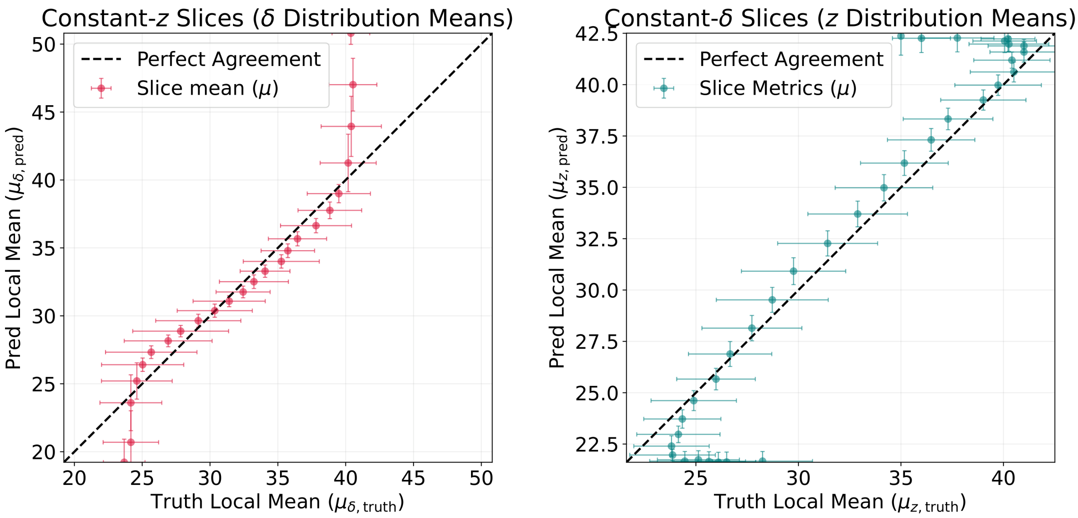

Fig. 9: Illustrative marginal reconstructions from the global causal Transformer. Coherent residual lobes indicate systematic displacement or deformation rather than zero-centred pixel noise.

For a marginal $H(u,v)$, the conditional mean of v at fixed u is

$$m_{v|u}(u) = \frac{\int vH(u,v)\,\mathrm{d}v}{\int H(u,v)\,\mathrm{d}v}. \tag{64}$$

This quantity probes internal dependence that may be hidden by global moments.

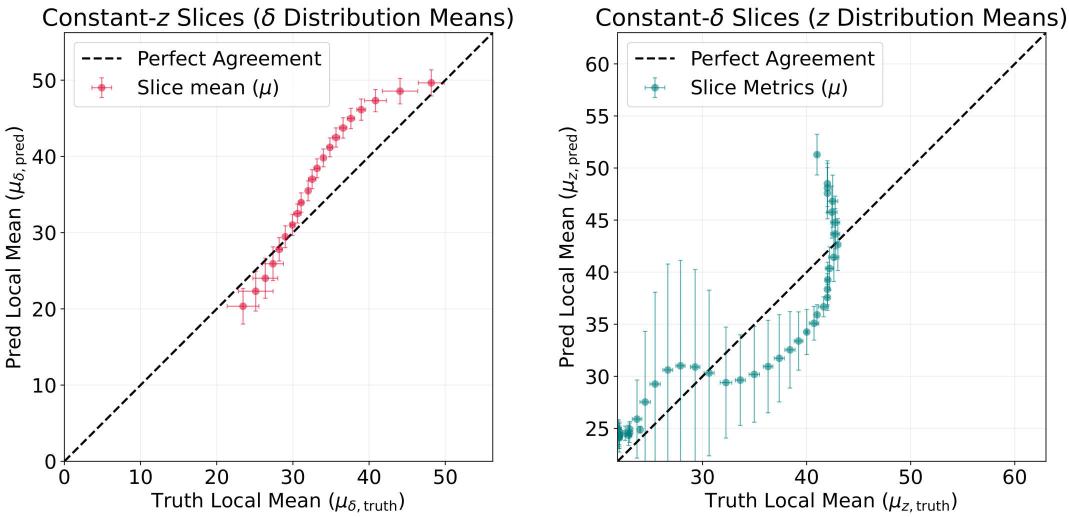

Fig. 10: Conditional-mean diagnostic for the global Transformer. Departures from the parity line indicate local phase-space dependence that is not reproduced by the prediction.

For reference marginal $H_p^{(s)}$ and prediction $\widehat{H}_p^{(s)}$, the signed residual is

$$\varepsilon_p^{(s)}(u,v) = \widehat{H}_p^{(s)}(u,v) - H_p^{(s)}(u,v). \tag{65}$$

Its test-set mean contributes to the bias norm B_H of Eq. (58). These diagnostics motivate the restricted-support comparison, but they do not by themselves identify global attention as the unique cause of the residuals.

5.4.2 Windowed attention and GNO

The same longitudinal conditional diagnostic is applied to the $W = 8$ Transformer and the GNO for pair $p = 11$, corresponding to (ζ, δ) under Eq. (32).

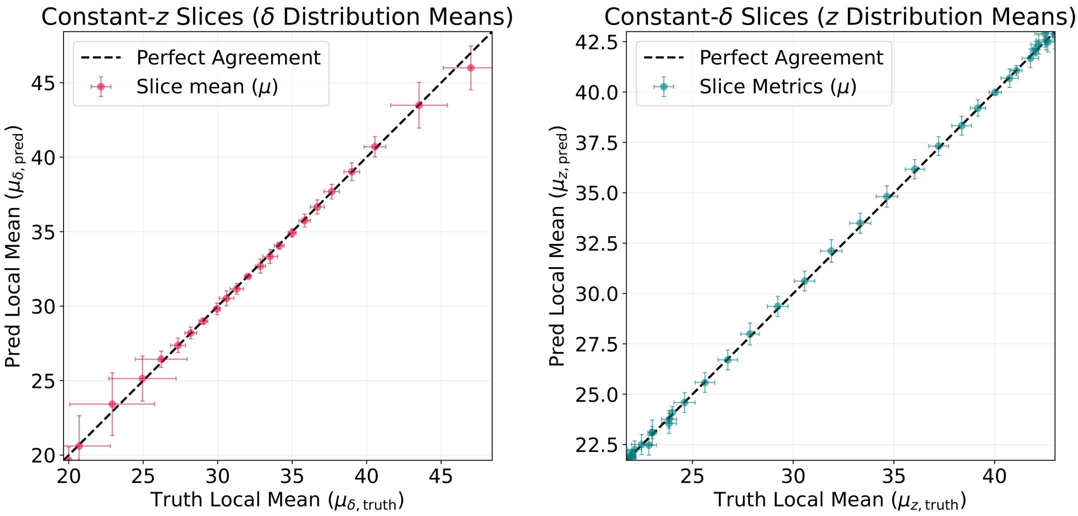

Fig. 11: Conditional-slice consistency of the $W = 8$ Transformer for the longitudinal (ζ, δ) marginal. Error bars encode the local conditional spread.

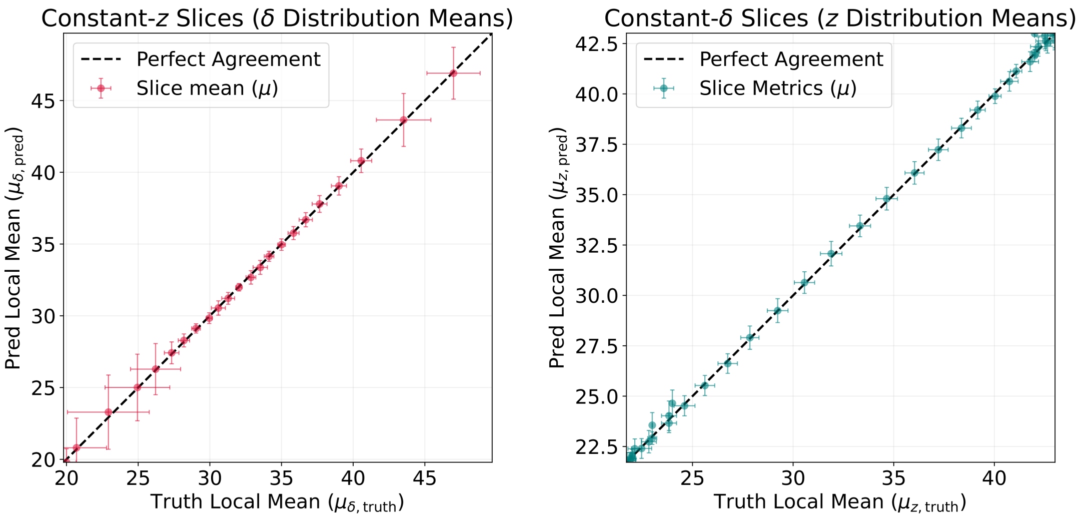

Fig. 12: Conditional-slice consistency of the CVAE–GNO surrogate for the longitudinal (ζ, δ) marginal. Comparison with Fig. 11 is descriptive; aggregate ranking uses Table 2.

The restricted-support models improve several aggregate metrics relative to global attention, but this comparison does not isolate locality in the strict causal sense because the configurations were independently optimised and not repeated across matched random seeds.

5.5 Selected marginal comparisons

Four planes from test sample $s = 3$ are used as qualitative examples:

$$p \in \{0, 1, 4, 11\} \quad \Longleftrightarrow \quad (x, y),\ (x, \zeta),\ (x, \delta),\ (\zeta, \delta). \tag{66}$$

The sample is not used to rank the models and should not be interpreted as a statistically representative median case. It illustrates how the aggregate errors appear in concrete projected densities.

5.5.1 Derived macroparticle visualisation

The predicted two-dimensional maps are additionally sampled to produce 10^6 synthetic coordinate pairs per plane. Fixed histogram edges determined from the training subset are used to map bins back to physical coordinates; test-target bounds are not recomputed. The resulting samples are not an independently predicted particle cloud. They contain the same information as the corresponding 2D histogram, add Monte Carlo sampling noise, and cannot be combined into a consistent six-dimensional ensemble.

5.6 Window-size dependence

The validation study compares causal windows $W \in \{1, 2, 4, 8, 16, 32\}$. For projection p,

$$E_p(W) = \frac{\left\| \widehat{H}_p^{(W)} - H_p \right\|_2}{\|H_p\|_2 + \varepsilon}, \tag{67}$$

and the reported scores are the mean and maximum over the fifteen channels.

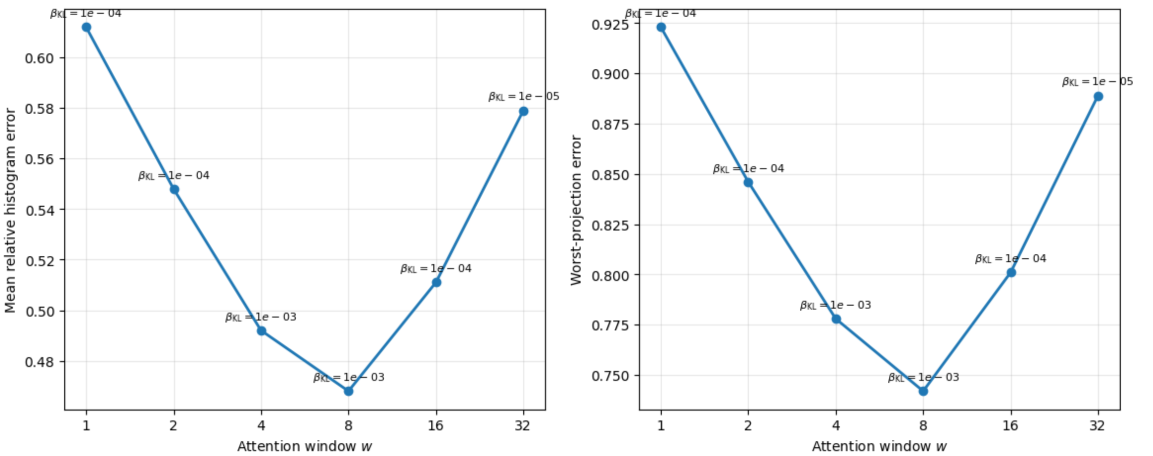

Fig. 17: Validation error as a function of causal window size. The KL weight was selected independently for each window, so the figure represents a tuned-model comparison rather than a fixed-regularisation ablation.

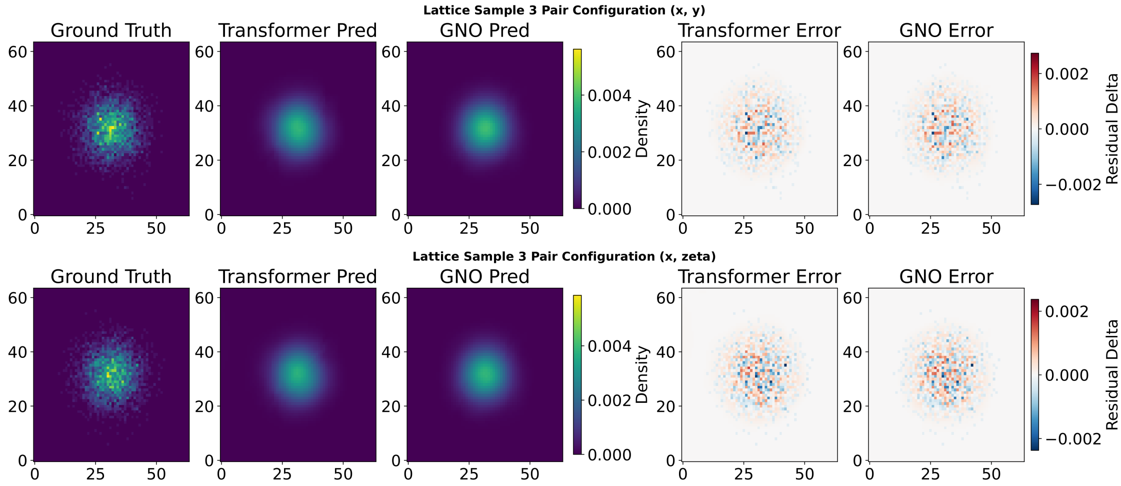

Fig. 13: Density-level comparison for illustrative sample $s = 3$. Top: (x, y), $p = 0$. Bottom: (x, ζ), $p = 1$. Each row contains the reference, $W = 8$ Transformer, GNO, and signed residual maps.

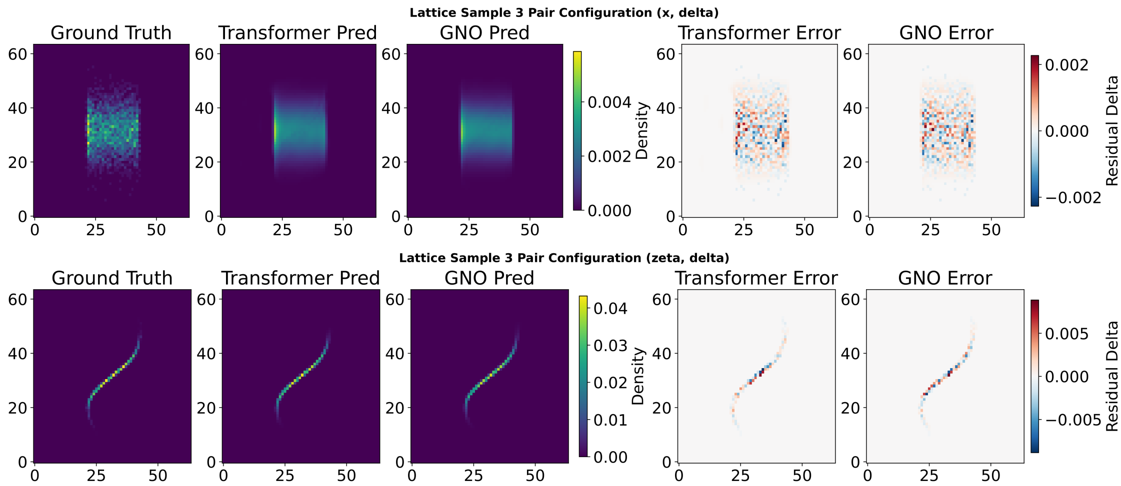

Fig. 14: Density-level comparison for illustrative sample $s = 3$. Top: (x, δ), $p = 4$. Bottom: (ζ, δ), $p = 11$.

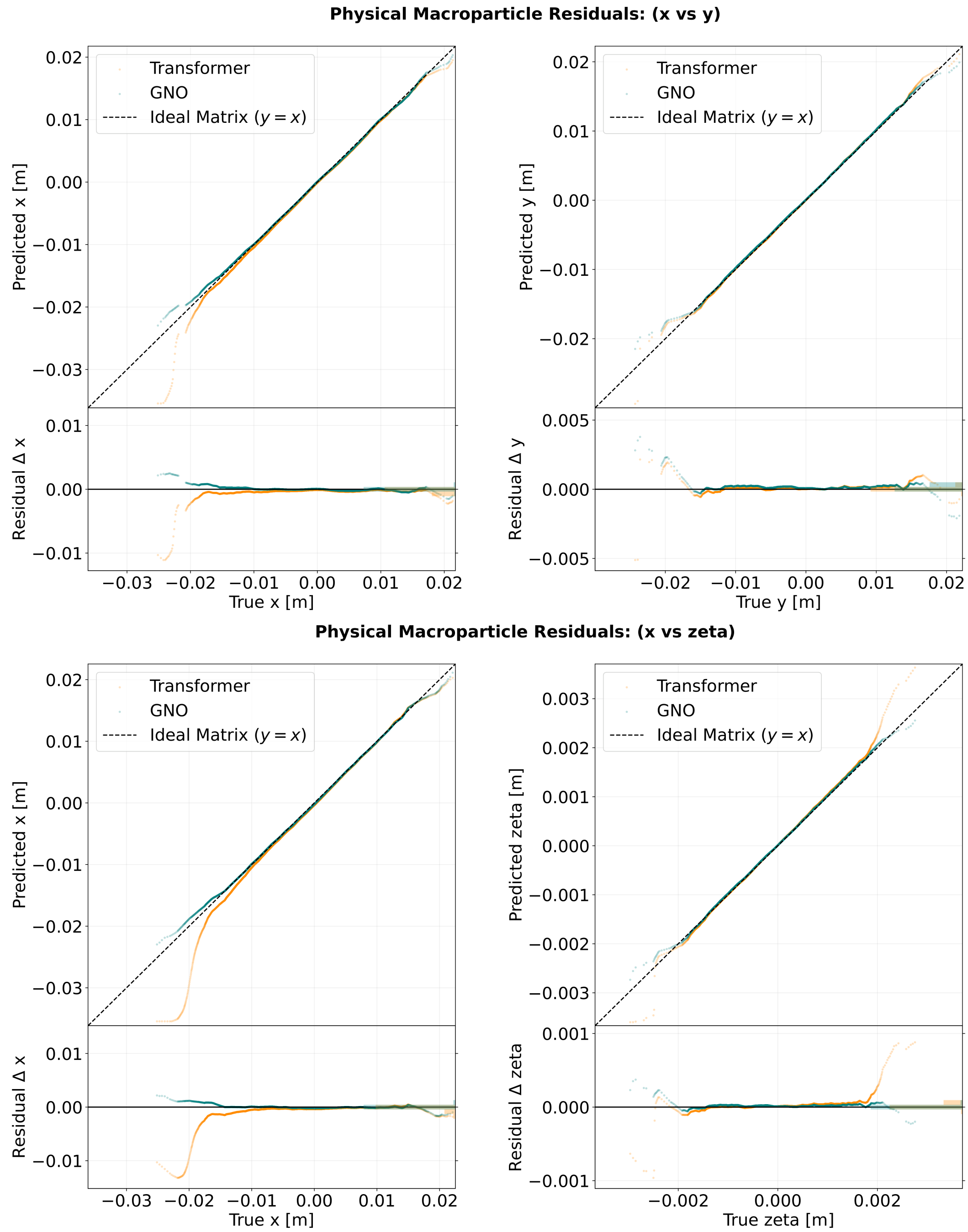

Fig. 15: Derived 2D sampling visualisation for (x, y) and (x, ζ) in sample $s = 3$. The plots translate density-map differences into coordinate-displacement diagnostics but provide no independent six-dimensional validation.

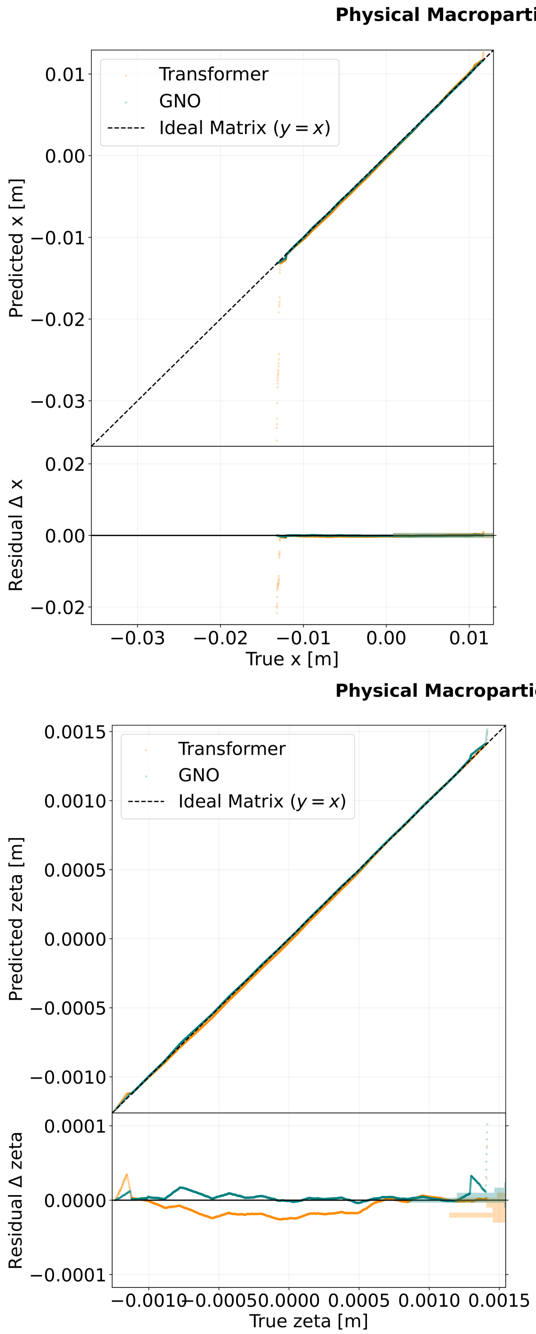

Fig. 16: Derived 2D sampling visualisation for (x, δ) and (ζ, δ) in sample $s = 3$.

Tab. 5: Validation-selected Transformer configurations. Each row corresponds to one selected checkpoint; uncertainty over training seeds is not available.

W	β_{KL}	Best epoch	Mean rel. error	Worst projection	Mean KL
1	1×10^{-4}	18	0.612	0.923	0.418
2	1×10^{-4}	24	0.548	0.846	0.356
4	1×10^{-3}	31	0.492	0.778	0.238
8	$\mathbf{1 \times 10^{-3}}$	**34**	**0.468**	**0.742**	**0.211**
16	1×10^{-4}	27	0.511	0.801	0.304
32	1×10^{-5}	21	0.579	0.889	0.447

The tuned validation scores are non-monotonic and attain their minimum at $W = 8$. This supports the practical use of an intermediate window in the present configuration. It does not, however, prove that $W = 8$ is a physical interaction length: β_{KL}, checkpoint epoch, and random optimisation trajectory also vary. A stricter ablation would repeat every W with fixed regularisation, matched model size, and several seeds.

5.7 Physical-statistics prediction

The statistics head predicts six centroids and six RMS sizes. Figure 18 shows parity plots, while the aggregate moment errors appear in Table 2.

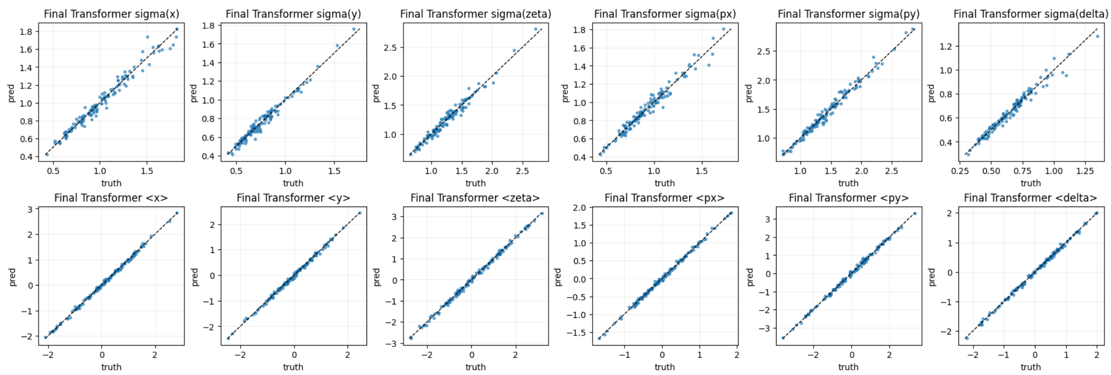

Fig. 18: Parity plots for predicted centroids and RMS sizes across the six coordinates. The plot reveals coordinate-dependent performance; agreement in RMS size does not imply equally accurate centroids.

The GNO has the lowest aggregate moment error, 0.547, whereas the Transformer heads are substantially less accurate. In particular, the parity plots indicate that centroid prediction is more difficult than RMS-size prediction. Coordinate-wise R_d^2, MAE, and bias values were not retained in the exported result bundle and are therefore not invented here. Consequently, Fig. 18 supports a qualitative coordinate-wise diagnosis, while the quantitative claim is restricted to the aggregate moment errors in Table 2. A final rerun should export the metrics defined in Eq. (61).

5.8 Result synthesis

Relative to global attention, both restricted-support approaches reduce ϵ_{rel} and SW in the selected test checkpoints. The $W = 8$ Transformer provides the lowest SW, while the GNO provides the best relative histogram, moment, bias, and runtime results. The data therefore support the narrower statement that structured support is beneficial for the evaluated models and split. They do not establish strict physical locality, universal architectural superiority, statistical significance across training runs, or operational validity outside the simulated parameter domain.

6 Discussion and perspectives

6.1 What the architectural comparison establishes

The study compares several hypotheses about how a reduced beam representation should evolve. The FNO uses a spectral parameterisation for a field defined on a regular longitudinal grid. The global Transformer treats the lattice as an ordered sequence with complete upstream access. The windowed Transformer replaces the complete causal graph by a banded graph, and the GNO uses explicit directed connectivity and a learned graph kernel. These models should therefore be understood as different structural approximations, not merely as networks of different sizes.

On the selected held-out checkpoints, global attention gives a relative histogram error of 0.681, whereas the $W = 8$ Transformer and GNO obtain 0.534 and 0.511, respectively. The GNO also has the smallest aggregate moment error, residual-bias norm, and inference time. However, the $W = 8$ Transformer has the smallest sliced-Wasserstein value, 0.014 compared with 0.022 for the GNO. The result is therefore a Pareto trade-off rather than a universal ranking.

The restricted-support models improve the chosen test metrics, but several explanations remain possible. A sparse interaction graph can act as statistical regularisation, reduce weakly constrained correlations, simplify optimisation, or alter effective capacity. Moreover, each window used an independently selected KL weight and checkpoint, and the comparison was not repeated over matched random seeds. The observed improvement is consequently evidence that structured support is a useful inductive bias for this dataset; it is not a controlled proof that locality alone causes the improvement.

The physical interpretation also requires care. Lattice transport is composed sequentially, but collective effects such as wakefields are non-local and directed. The token neighbourhood of a latent propagator is not the same object as the longitudinal head–tail support of a wake function. Windowed attention and graph message passing are computationally related through restricted aggregation, yet neither kernel is identified with a physical wake or Green function. Establishing such an interpretation would require dedicated kernel diagnostics and perturbations of the lattice or bunch ordering.

6.2 Representation limits and physical consistency

The fifteen pairwise marginals provide a tractable reduction from particle clouds to $15 \times 64 \times 64$ arrays. They retain non-Gaussian projected structure, including tilt, halo, asymmetry, and multimodality, while remaining compatible with convolutional encoders. The price of this reduction is non-identifiability: distinct six-dimensional joint densities can possess the same complete set of pairwise marginals. The model therefore reconstructs a pairwise-marginal representation of a six-dimensional beam state, not a unique six-dimensional density.

Independent channel normalisation preserves unit mass in every projected density but does not guarantee that two channels sharing a coordinate yield the same one-dimensional marginal. Equation (35) provides a direct diagnostic for this defect. A natural extension is to add

$$\mathcal{L}_{\text{cons}} = \lambda_{\text{cons}} \mathcal{E}_{\text{cons}} \tag{68}$$

to the reconstruction objective or to construct the decoder around shared one-dimensional marginal factors. Such a constraint would connect probabilistic consistency to the architecture more directly than generic regularisation.

The explicit statistics head addresses a different limitation. A normalised histogram can reproduce morphology while missing absolute displacement or extension. Predicting six centroids and six RMS sizes in parallel separates these quantities from projected shape. The current results indicate that RMS extensions are easier to predict than small centroid shifts. Future models should standardise each coordinate by a train-only physical scale and report coordinate-wise R^2, MAE, and signed bias.

6.3 Prospective uncertainty quantification

The stochastic CVAE posterior permits repeated latent samples,

$$z^{(r)} \sim q_\phi(z \mid X, a, d), \qquad r = 1, \dots, R, \tag{69}$$

which can be propagated to form an ensemble $\widehat{H}^{(r)}$. The corresponding sample mean and variance describe sensitivity to the encoded latent distribution. They must not automatically be interpreted as calibrated predictive uncertainty. Posterior sampling mainly reflects the learned conditional latent variability; it does not by itself quantify uncertainty in network parameters, simulator discrepancy, or out-of-distribution extrapolation.

A calibrated study would separate at least three contributions: latent or aleatoric variability, epistemic variability across independently trained models, and reference-simulation uncertainty. Calibration should be evaluated through empirical coverage, reliability curves, proper scoring rules, and the relation between predicted variance and observed error. Until those tests are performed, uncertainty quantification remains a perspective rather than a demonstrated result.

6.4 Toward optimisation and accelerator digital twins

Once validated, a differentiable surrogate can support sensitivity analysis, constrained parameter scans, and Bayesian optimisation [4]. The present models are not yet an operational digital twin: they have not been calibrated to machine measurements, tested under distribution shift, or coupled to an online state-estimation loop. They should instead be regarded as components of a possible digital-twin workflow in which high-fidelity simulation, measurement, uncertainty estimates, and a fast surrogate are updated together.

6.5 Limitations

The conclusions of this report are subject to the following limitations:

1. the principal six-dimensional benchmark contains 512 simulated trajectory pairs and one fixed train–validation–test split;
2. the selected propagator checkpoints were not evaluated over several matched random seeds, so optimisation variability and statistical significance are unknown;
3. the window study changes both attention width and the independently selected KL weight, and is therefore not a fixed-regularisation ablation;
4. the pairwise representation does not uniquely determine the full six-dimensional joint density and may violate consistency among shared one-dimensional marginals;
5. the CVAE is lossy, with a test relative reconstruction error of 0.367, so downstream errors combine compression and propagation effects;
6. the latent Pearson diagnostic uses only 51 validation samples and lacks permutation, multiple-testing, and nonlinear-dependence controls;
7. the macroparticle dashboards resample two-dimensional predicted maps and do not constitute an independently generated six-dimensional particle cloud;
8. coordinate-wise R^2, MAE, and bias were not retained for the statistics head, preventing a complete quantitative assessment of centroid and RMS predictions;
9. reference-simulation runtimes were not measured under a common protocol, so no simulator speed-up is claimed;
10. the models do not exactly enforce symplecticity, detailed wakefield causality, cross-marginal consistency, or calibrated uncertainty;
11. all quantitative conclusions are in-distribution and simulation-based; extrapolation to unseen machine settings or operational FCC-ee data has not been demonstrated.

These limitations define a concrete validation programme. It comprises fixed-hyperparameter and repeated-seed ablations, stronger baselines, out-of-distribution splits, shared-marginal consistency, coordinate-wise physical metrics, calibrated ensembles, and end-to-end timing against identified high-fidelity simulations.

7 Conclusion

This internship investigated simulation-based machine-learning surrogates for collective-effects studies relevant to the FCC-ee High-Energy Booster. Two modelling levels were considered. A Fourier Neural Operator was first evaluated on longitudinal density maps. Separately, six-dimensional particle clouds were reduced to fifteen pairwise phase-space marginals, six centroids, and six RMS sizes. A conditional variational autoencoder compressed the marginal representation, after which global causal attention, finite-window attention, and a Graph Neural Operator propagated the latent beam state.

The central empirical result is that restricting the support of latent information transfer improves the evaluated models relative to global causal attention. On the common held-out test set, the relative histogram error decreases from 0.681 for the global Transformer to 0.534 for the $W = 8$ Transformer and 0.511 for the GNO. The $W = 8$ Transformer obtains the lowest sliced-Wasserstein discrepancy, 0.014, whereas the GNO obtains the lowest aggregate moment error, 0.547, residual-bias norm, 0.376, and inference time, approximately 2.34 ms per sample. The comparison therefore identifies complementary strengths rather than a single model that dominates every criterion.

The work also clarifies what these results do not establish. The fifteen marginals are a reduced representation and do not uniquely reconstruct the complete six-dimensional joint density. The reported checkpoints come from one data split without matched repeated seeds, reference-simulation costs were not measured under a common protocol, and coordinate-wise physical-statistics scores were not retained. The results consequently support the conclusion that structured causal support and graph connectivity are useful inductive biases within the sampled simulation regime; they do not prove strict physical locality, calibrated uncertainty, operational speed-up, or validity outside the training domain.

The most direct next steps are to enforce consistency among pairwise marginals and to export coordinate-wise R^2 and bias. The architectural comparison should then be repeated with fixed regularisation, multiple seeds, and out-of-distribution parameter splits. End-to-end simulation and inference costs must also be measured on identified hardware. With these validations, latent operator models could become useful components of uncertainty-aware parameter scans and future accelerator digital-twin workflows.

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A Theoretical scope and prospective inverse problem

A.1 What the cited approximation results establish

The neural-operator results used to motivate the methodology are approximation statements for maps between function spaces, not guarantees about a particular trained checkpoint. Under the hypotheses of Theorem 8 in [7], a fixed continuum neural operator has finite-dimensional realisations that converge under mesh refinement, uniformly on compact subsets of the admissible input space. Schematically, for fixed parameters θ,

$$\sup_{a\in K} \left\| \widehat{G}_{L,\theta}(D_L, a|_{D_L}) - G_\theta(a) \right\|_U \longrightarrow 0 \qquad (L \to \infty). \tag{70}$$

This is a statement about consistency of the discretised architecture. It does not establish robustness to changing the number of Monte Carlo macroparticles used to estimate an input density.

Likewise, Theorems 11–14 in [7] provide density results for neural operators under specified domain, function-space, continuity, activation, and measurability hypotheses. On a compact admissible set K, the uniform-approximation implication has the form

$$\forall \varepsilon > 0, \quad \exists G_\theta : \sup_{a\in K} \left\| G^\dagger(a) - G_\theta(a) \right\|_U < \varepsilon. \tag{71}$$

Related FNO-specific approximation results are given in [8]. These statements are existential: they do not prove that the finite width, retained Fourier modes, optimiser, sample size, or training budget used in this report attain a prescribed error. Because the reported FNO experiment was evaluated on its training-grid resolution, this thesis makes no empirical claim of cross-resolution generalisation.

Earlier drafts discussed Hénon-type symplectic networks. They are not presented as a contribution or guarantee here because no such architecture was trained or evaluated in the reported experiments. Structure preservation is therefore kept as a possible future direction rather than evidence for the present surrogate.

A.2 Conditional variational objective

Let X be the fifteen-channel beam representation, a its physical summary, d the input/output-domain indicator, and z the latent variable. The implemented encoder is $q_\phi(z \mid X, a, d)$, the decoder represents $p_\theta(X \mid z, d)$, and the prior is $p(z) = \mathcal{N}(0, I)$. Since a is computed from the same beam state, the variational posterior remains an admissible function of the observation. The conditional evidence decomposes as

$$\log p_\theta(X \mid d) = \mathcal{L}_{\mathrm{ELBO}}(X, a, d) \tag{72}$$

$$+ D_{\mathrm{KL}}(q_\phi(z \mid X, a, d) \,\|\, p_\theta(z \mid X, d)), \tag{73}$$

where

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi}\left[\log p_\theta(X \mid z, d)\right] - D_{\text{KL}}(q_\phi(z \mid X, a, d) \parallel p(z)) \,. \tag{74}$$

Non-negativity of the final KL divergence gives $\mathcal{L}_{\text{ELBO}} \leq \log p_\theta(X \mid d)$ [9, 10]. The bound is tight only when the variational posterior equals the exact posterior almost everywhere. The centroid and RMS penalties in Eq. (48) are additional supervised regularisers, so the complete implemented objective is not the negative ELBO alone. Neither the identity nor the added losses guarantee latent identifiability, calibrated uncertainty, or absence of posterior collapse.

A.3 Exact support constraints of the latent propagators

For the windowed Transformer, W counts the accessible tokens including the current token. Its one-layer upstream neighbourhood is

$$\mathcal{N}_W^-(i) = \{j : \max(1, i - W + 1) \leq j \leq i\} \,. \tag{75}$$

In the absence of a global bypass, a masked-attention layer satisfies

$$\frac{\partial h_i^{(\ell+1)}}{\partial h_j^{(\ell)}} = 0, \qquad j \notin \mathcal{N}_W^-(i). \tag{76}$$

Thus $W = 1$ is self-only attention. After L stacked layers, indirect information transfer may span at most $L(W-1)$ upstream index steps, or $1 + L(W-1)$ token positions, subject to boundary effects and other paths in the architecture. The mask therefore enforces local support per layer, not strict locality of the complete end-to-end map.

For a GNO layer, node j contributes directly to node i only if $j \in \mathcal{N}_G(i)$:

$$v_i^{(\ell+1)} = \Psi_\ell\left(v_i^{(\ell)}, \sum_{j \in \mathcal{N}_G(i)} \omega_{ij} K_{\theta,\ell}(h_i, h_j, v_i^{(\ell)}, v_j^{(\ell)}, \mu) v_j^{(\ell)}\right) \,. \tag{77}$$

Repeated graph layers expand the receptive field through graph hops. This exact computational sparsity motivates the architectural comparison, but it does not prove that the physical wakefield or collective interaction has compact support in lattice index.

A.4 Approximation and discretisation errors

Let $G^\dagger$ be the target physical operator, G_θ the learned continuum operator, and $\widehat{G}_{L,\theta}$ its discrete realisation. The triangle inequality separates two sources of error:

$$\begin{aligned} &\left\|\widehat{G}_{L,\theta}(D_L, a|_{D_L}) - G^\dagger(a)\right\|_U \\ &\quad \leq \underbrace{\left\|\widehat{G}_{L,\theta}(D_L, a|_{D_L}) - G_\theta(a)\right\|_U}_{\text{discretisation error}} + \underbrace{\left\|G_\theta(a) - G^\dagger(a)\right\|_U}_{\text{operator-approximation error}} \,. \end{aligned} \tag{78}$$

Discretisation invariance addresses the first term for fixed parameters under mesh refinement; universal approximation concerns the representational capacity associated with the second term. Neither statement controls optimisation error, finite-sample estimation error, distribution shift, or autoregressive accumulation. Those empirical limitations are listed explicitly in Chapter 6.

A.5 Prospective wake reconstruction from Haïssinski profiles

The following inverse problem is a proposed extension and was not implemented or evaluated in the reported experiments. Under a chosen nondimensionalisation and sign convention, let $q = z/\sigma_{z0}$ and let $\lambda(q) > 0$ be a normalised Haïssinski profile satisfying

$$\frac{\mathrm{d}\lambda}{\mathrm{d}q} + [q - F(q;\lambda)]\,\lambda(q) = 0, \qquad F(q;\lambda) = I\int_{-\infty}^{\infty} W(q-q')\lambda(q')\mathrm{d}q'. \tag{79}$$

Then

$$F(q) = \frac{\mathrm{d}}{\mathrm{d}q}\log\lambda(q) + q. \tag{80}$$

Profiles measured or simulated at several currents lead to an inverse convolution problem [16, 17]. On a uniform grid, one possible regularised estimator is

$$\widehat{w} = \arg\min_{w}\left\{\frac{1}{M}\sum_{n=1}^{M}\left\|\widehat{\boldsymbol{F}}^{(n)}(w) - \boldsymbol{F}^{(n)}\right\|_2^2 + \alpha\left\|D^2 w\right\|_2^2 + \beta\|w\|_2^2\right\}. \tag{81}$$

Because differentiating $\log\lambda$ amplifies sampling and measurement noise, any future study must specify smoothing, support, causality, current calibration, and regularisation before interpreting $\widehat{w}$ as a reconstructed physical wake.